When Does Test-Time Ensembling Help Large Language Models? A Budget-Matched Study of Answer-Space and Prompt-Space Diversity

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Abstract

Test-time ensembling improves the reliability of large language models by aggregating answers from multiple model calls instead of trusting a single greedy prediction. Two families of strategies dominate practice. Self-consistency resamples the stochastic decoder on one fixed prompt, so the ensemble varies along what we call the answer space. Prompt ensembling instead fixes the decoding configuration and varies the prompt over semantically equivalent paraphrases, so the ensemble varies along the prompt space. Both families consume the same scarce resource, namely the number of model calls, yet prior work studies them in isolation and the question of which source of diversity buys more accuracy per call remains open. We present a controlled, budget-matched comparison across six commercial large language models and six reasoning and knowledge benchmarks under a unified protocol in which every question is answered sixteen times per model: six decodes of the original prompt, six paraphrased prompts answered once each, a hybrid allocation that combines three paraphrases with two decodes each, a twelve-vote combined condition, and a greedy baseline, together with a single-sample baseline, a greedy-paraphrase control that isolates prompt-space effects from decoding stochasticity, a chain-of-thought variant, and a hard-task probe on MMLU-Pro. The results reveal a regime structure that current evaluation practice overlooks. On benchmarks that modern flagship models saturate, the sampled answer distribution degenerates to a single

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value, both ensembles collapse onto the greedy baseline, and consistency-based uncertainty estimates lose their discriminative power, which raises a caution for sampling-based confidence in this regime. Where genuine answer entropy exists, both ensembles lift accuracy over greedy decoding, no significant difference between the two diversity sources is detected at matched budgets (though the study is not powered to establish equivalence), a single stochastic draw can differ from greedy decoding by ten points, majority voting can even reduce accuracy when the modal answer is wrong, and the two agreement signals flag partially disjoint errors across models. We distill the findings into an adaptive allocation rule that matches or exceeds every fixed strategy in simulation at two-thirds of the budget, and we release all prompts, paraphrases, and raw predictions to support reproducible evaluation.

Keywords: Large language models, test-time compute, self-consistency, prompt ensembling, uncertainty estimation, robustness

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1. Introduction

Large language models answer questions through a single forward pass whose output depends on two design choices that the user controls: the prompt that specifies the task, and the decoding configuration that maps model probabilities into text. Both choices turn out to be fragile. A long line of work on prompt sensitivity shows that semantically equivalent reformulations of the same question can change model answers by large margins [1, 2, 3, 4]. A parallel line of work on test-time compute shows that a single greedy decode is an unreliable summary of what the model knows, and that aggregating several decodes of the same prompt, as in self-consistency [5], improves reasoning accuracy at the cost of additional inference calls.

These two observations motivate two different ways of spending a fixed budget of model calls. The first spends the budget on the answer space: keep the prompt fixed, sample several decodes at nonzero temperature, and aggregate them by majority vote. The second spends the budget on the prompt space: keep decoding deterministic or lightly stochastic, present several semantically equivalent paraphrases of the task, and aggregate the resulting answers. Both strategies are used in practice, both appear in the evaluation literature under separate headings, and both consume the same scarce resource. Surprisingly, the literature offers little guidance on which source of diversity yields more accuracy per call, whether the two sources detect the

same errors, and how the answer changes with model capability and task difficulty.

The question is becoming more urgent as model behavior shifts. Post-trained flagship models are increasingly confident: on many legacy benchmarks they answer near ceiling, and preliminary reports suggest that repeated sampling at moderate temperature often returns identical answers, which would render answer-space ensembling inert precisely where deployments are most common. At the same time, uncertainty estimation methods built on sampling, such as semantic entropy [6, 7], implicitly assume that the sample distribution carries information about correctness. If that assumption fails for confident models, practitioners need to know before they build on it.

This paper presents, to our knowledge, the first budget-matched, within-question comparison of answer-space and prompt-space ensembling across multiple model families and task types. Every question in our protocol is answered sixteen times by every model under matched conditions: one greedy decode, six stochastic decodes of the original prompt, one stochastic decode of each of six validated paraphrases, and three additional second decodes that support a hybrid condition. From this single call matrix we derive self-consistency ensembles of any size up to six, paraphrase ensembles of the same sizes, a hybrid allocation, a combined twelve-vote ensemble, per-question consistency scores for both diversity sources, and fragility measures that record how often paraphrasing alone flips the majority answer. Three follow-up conditions then isolate the factors the matrix alone cannot separate: a greedy-paraphrase control that removes decoding stochasticity from the paraphrase side, a chain-of-thought variant that changes the answering mode, and a hard-task probe on MMLU-Pro. Because all conditions are derived from matched calls, comparisons between strategies hold the question, the model, and the total call budget constant by construction.

Our study addresses four research questions. First, under equal budgets, does spending calls on paraphrases buy more, equal, or less accuracy than spending them on resampling, and does the hybrid allocation dominate either pure strategy? Second, how brittle are model answers to paraphrasing per se, and where do paraphrase-induced flips concentrate? Third, can agreement across paraphrases serve as an error signal in the way that agreement across samples does, and how do the two signals compare when both are available? Fourth, how do all of these effects interact with model capability and task difficulty? The last question turns out to govern the first three.

The empirical picture that emerges has a clear regime structure. When a model finds a task easy, its answer distribution collapses: across our six benchmarks, flagship models return the same answer for essentially every sample and every paraphrase on saturated tasks, self-consistency and paraphrase ensembling both collapse onto the greedy baseline, and consistency scores carry no signal about which answers are wrong. When a task genuinely challenges a model, answer entropy reappears, both ensemble families improve over greedy decoding, the two consistency measures flag partially disjoint errors, and a hybrid allocation protects against a failing diversity source. This structure implies that no single published claim about ensembling generalizes across regimes, and that practical systems should first measure the cheap statistics that identify the regime before choosing how to spend their budget.

Our contributions are as follows. We formalize test-time ensembling as a choice between two diversity sources under a shared budget, and we introduce a call matrix design that derives all conditions of interest from one matched set of calls, with controls that separate prompt-space effects from decoding stochasticity, answering mode, and task difficulty. We provide a multi-model, multi-benchmark empirical map of when each source helps, including systematic evidence that paraphrase agreement functions as an uncertainty signal whose error coverage only partially overlaps that of sample agreement. We document the answer degeneracy phenomenon that renders both ensembling and consistency-based uncertainty estimation inert on saturated tasks for confident models, with direct implications for sampling-based confidence estimation in this regime. Finally, we translate the findings into an adaptive allocation rule validated by simulation, and we release our complete prompt materials, validated paraphrases, and per-call predictions.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 formalizes the ensembling framework and the protocol. Section 4 describes models, benchmarks, and validation procedures. Section 5 presents the results around our four research questions. Section 6 discusses implications and limitations, and Section 7 concludes.

2. Related Work

2.1. Test-time compute and self-consistency

Scaling computation at inference time has become a standard tool for improving the reliability of large language models. Chain-of-thought prompting

demonstrates that eliciting intermediate reasoning steps raises accuracy on arithmetic and symbolic tasks [8], and self-consistency extends this observation by sampling multiple reasoning paths at nonzero temperature and returning the most frequent final answer [5]. Follow-up work reduces the number of samples needed through early stopping [9], generalizes aggregation to free-form generation by letting the model itself select among candidate answers [10], and studies the optimal allocation of test-time compute relative to model size [11]. Repeated sampling studies document that coverage grows log-linearly with the sample budget but that majority voting plateaus without a verifier [12], a plateau our regime analysis explains: where the modal answer is wrong, additional votes amplify the error. Process reward models that score intermediate steps represent a further refinement of the same budget idea [13]. Multi-agent debate aggregates answers across multiple model instances and rounds [14], with prompt variation proposed as one axis of diversity of thought [15], and controlled comparisons find that debate does not reliably outperform self-consistency at matched budgets [16], which reinforces the need for budget-matched comparisons when evaluating aggregation strategies. Our work differs from this literature in the source of diversity it studies: instead of asking how many samples to draw or how to weight them, we ask whether the same calls are better spent perturbing the prompt, and we answer under a matched-budget protocol.

2.2. Prompt sensitivity and prompt ensembling

A separate literature documents that model outputs are sensitive to semantically irrelevant details of the prompt. Reformatting alone, such as changing spacing or option ordering, shifts accuracy by several points [2], and the measured performance of a model depends strongly on which of many equivalent prompts an evaluator happens to choose [1]. Linguistic properties of the prompt, such as lexical familiarity, partially predict its success [3], and adversarial prompt perturbations can degrade performance further [4]. The natural constructive response is to aggregate over prompts. Ask Me Anything generates many question reformulations with a language model and combines the resulting pseudo-labels [17], an idea that predates and parallels self-consistency but targets zero-shot classification rather than matched-budget reasoning. Boosted prompt ensembles construct few-shot prompt sets stepwise on hard examples and outperform single-prompt output-space ensembles on GSM8K and AQuA [18]; diversity-of-thought prompting varies the prompt to induce diverse reasoning paths before aggregation [15]; and evalua-

tion studies propose averaging over prompt templates to stabilize benchmark scores [1]. These methods share our premise that the prompt is a legitimate axis of ensembling, but none of them isolates prompt-space diversity from decoding stochasticity under a matched budget, and none quantifies when prompt-space aggregation beats answer-space aggregation on identical questions, which is the gap our protocol fills.

2.3. Uncertainty estimation from sampled behavior

Between the two aggregation literatures sits a third that uses sampling not to improve accuracy but to estimate uncertainty. Verbalized confidence asks the model to state a probability [19, 20], and self-probing methods ask the model to evaluate its own candidate answers [21]. Behavioral methods infer uncertainty from the dispersion of sampled outputs, from simple answer agreement to semantic clustering that treats paraphrased answers as equivalent [6, 7], with surveys consolidating the rapidly growing set of measures [22]. Perturbation-based methods broaden the probe beyond decoding: SPUQ perturbs the input, samples outputs per perturbation, and aggregates [23], and input clarification ensembling decomposes LLM uncertainty into data and model components by generating input clarifications and ensembling the predictions [24]. Calibration metrics such as expected calibration error [25] and selective prediction metrics such as the area under the risk-coverage curve quantify how well a confidence score ranks correct above incorrect answers. These methods inherit an assumption from sampling: that the sample distribution is informative. Our study tests this assumption directly, characterizes the regimes in which sample agreement and paraphrase agreement carry signal, and quantifies how much of the two signals' error coverage overlaps. In this sense our fragility analysis also connects to work on the faithfulness of chain-of-thought explanations, which finds that stated reasoning does not always cause the final answer [26]: paraphrase stability measures the robustness of the answer to surface form rather than to the reasoning trace.

2.4. Positioning

In summary, answer-space ensembling, prompt-space ensembling, and sampling-based uncertainty estimation have developed largely in isolation. Answer-space methods assume resampling generates useful diversity. Prompt-space methods assume paraphrases probe the same knowledge. Uncertainty methods assume sample dispersion tracks correctness. All three

assumptions are regime dependent, and no prior work, to our knowledge, measures all three under a single matched-budget protocol on modern models. The comparison also carries practical weight for evaluation practice, since benchmark conclusions about model capability shift when prompts vary [1] and when aggregation is applied [5]. Our contribution is the unified empirical map, the validation machinery for paraphrases, and the decision guidance that follows from the regime structure.

3. A Budget-Matched Framework for Test-Time Ensembling

3.1. Setup and notation

Let M denote a large language model and let q denote a question with a gold answer $y^*(q)$. For multiple-choice tasks the answer is an option letter, and for free-form numeric tasks the answer is a value compared numerically. A prompt x is a rendering of q into natural language, consisting of an optional passage, a question stem, an option list, and an instruction that specifies the required output format. We distinguish one canonical prompt x_0 and K paraphrased prompts $x_1, \ldots, x_K$ that preserve the semantics of x_0 under a validation procedure described in Section 4.

The model produces an answer through stochastic decoding. We write $a(M, x, s)$ for the answer extracted from the s-th independent decode of prompt x at temperature $\tau = 0.7$, and $a_g(M, x)$ for the greedy decode at $\tau = 0$. Answers are extracted from the model output by a fixed parser that searches for the mandated answer format and falls back to pattern matching over the final output lines; the parse rate is reported in Section 4, together with a retry policy and a sensitivity analysis for failed extractions.

3.2. The call matrix

For every question and model we collect a matrix of sixteen calls:

$$\mathcal{C}(M, q) \;=\; \big\{a_g(M, x_0)\big\} \;\cup\; \bigcup_{s=1}^{S} \big\{a(M, x_0, s)\big\} \;\cup\; \bigcup_{k=1}^{K} \big\{a(M, x_k, s) : s \le r_k\big\}, \tag{1}$$

with $S = 6$ samples on the canonical prompt, $K = 6$ paraphrases, and $r_k = 2$ decodes for the first three paraphrases and $r_k = 1$ for the remaining three. The matrix therefore contains one greedy call, six canonical samples, six first-decode paraphrase calls, and three second-decode paraphrase calls.

Every experimental condition studied in this paper is a deterministic function of this matrix, which guarantees that all comparisons hold the question, the model, and the realized decoding randomness constant across conditions.

3.3. Ensemble conditions

All conditions aggregate votes by majority rule. Ties are broken in favor of the earliest vote in a fixed canonical order, which prefers the canonical prompt over paraphrases and earlier paraphrases over later ones; we report tie incidence, which is negligible except where noted.

The *self-consistency* condition of size k aggregates the first k canonical samples:

$$\text{SC-}k(q) \;=\; \text{majority}\big(a(M, x_0, 1), \ldots, a(M, x_0, k)\big), \qquad k \le 6. \tag{2}$$

The *paraphrase ensemble* of size k aggregates one decode of each of the first k paraphrases:

$$\text{PE-}k(q) \;=\; \text{majority}\big(a(M, x_1, 1), \ldots, a(M, x_k, 1)\big), \qquad k \le 6. \tag{3}$$

Both conditions spend exactly k model calls and differ only in where the diversity comes from: SC-k perturbs the decoder while the prompt is fixed, whereas PE-k perturbs the prompt while the decoder configuration is fixed. This is the budget match that makes the two conditions comparable.

The *hybrid* condition splits a budget of six calls between the two sources, using two decodes of each of the first three paraphrases:

$$\text{HB-6}(q) \;=\; \text{majority}\big(a(M, x_k, s) : k \in \{1, 2, 3\},\ s \in \{1, 2\}\big). \tag{4}$$

Finally, the *combined* condition aggregates all twelve stochastic votes available in the matrix, $\text{COMB-12}(q) = \text{majority}\big(\text{SC-6}(q) \cup \text{PE-6}(q)\big)$, and serves as a twelve-call reference point for the marginal value of doubling the budget. Three further baselines and controls complete the design. The *single-sample* baseline S1 is one stochastic draw of the canonical prompt; its expected accuracy is estimated by averaging over the six canonical samples, and it differs from the greedy baseline whenever temperature-0.7 decoding and temperature-0 decoding disagree. A *greedy-paraphrase* control re-decodes each paraphrase at $\tau = 0$, giving $\text{PE}^g\text{-6}(q) = \text{majority}\big(a_g(M, x_1), \ldots, a_g(M, x_6)\big)$; comparing PE^g-6 with PE-6 isolates the contribution of prompt-space variation from the decoding stochasticity that the standard paraphrase ensemble mixes in. A

chain-of-thought variant repeats the single-call baseline, the self-consistency condition, and (for stem-locus datasets) the paraphrase ensemble under an explicit reasoning instruction, and a hard-task probe repeats the single-call and self-consistency conditions on MMLU-Pro; both are specified in Section 4 and analyzed in Section 5. The greedy decode $a_g(M, x_0)$ serves as the single-call baseline.

3.4. Consistency, entropy, and fragility

Three per-question statistics summarize the answer distribution. *Sample agreement* is the frequency of the modal answer among the six canonical samples,

$$c_{\mathrm{S}}(q) \;=\; \max_{v} \frac{1}{6} \sum_{s=1}^{6} \mathbb{1}\big[a(M, x_0, s) = v\big], \tag{5}$$

and *paraphrase agreement* $c_{\mathrm{P}}(q)$ is defined analogously over the six first-decode paraphrase answers. Both lie in $(0, 1]$, and the value 1 indicates that the corresponding source produced no diversity at all, a situation we call *answer degeneracy*. The *support-normalized answer entropy* is the Shannon entropy of the empirical answer distribution over the twelve stochastic votes, divided by the maximum entropy attainable with the observed number of distinct answers; it equals one when the observed answers are uniformly distributed over their support and zero when they coincide. We use this normalization throughout, and all entropy values reported in the paper use it.

Fragility captures disagreement between the two views of the same question: a question is fragile for a model when the majority answer of the canonical samples differs from the majority answer of the paraphrase answers, that is, when SC-6(q) $\neq$ PE-6(q). Fragility measures how much of the model's answer is an artifact of the particular surface form of the prompt rather than of the underlying question. The greedy-paraphrase control supports a noise-free variant, SC-6(q) $\neq$ PE^g-6(q), in which neither side involves stochastic decoding.

3.5. Error detection and selective prediction

We evaluate consistency scores as unsupervised error detectors. For each model, the consistency score of a question (reversed, so that lower agreement indicates higher predicted risk) ranks questions, and we compute the area under the receiver operating characteristic curve (AUROC) for predicting

whether the reference answer is incorrect, where the reference is the combined twelve-vote ensemble unless stated otherwise. When both agreement scores are available, the *combined* detector is their sum of disagreements, $2 - c_{\mathrm{S}} - c_{\mathrm{P}}$, so that a question is flagged when either source shows diversity; this is the synthesis used in the Both columns of Section 5. We complement AUROC with risk-coverage curves that plot the error rate of the accepted subset as the acceptance threshold varies, summarized by the area under the risk-coverage curve (AURC), computed by accepting questions in decreasing order of confidence and averaging the error rate over all prefix sizes; smaller AURC indicates a better ranking, and a random ranking yields AURC equal to the overall error rate. The oracle-excess AURC (E-AURC) measures the gap to the best achievable ranking [25, 22]. Where a model produces zero errors or constant scores, AUROC is undefined and we report the regime instead; where a model produces fewer than five errors, we report AUROC and AURC descriptively only, without confidence intervals or cross-model comparison.

Because the agreement scores and the reference answer share the same underlying votes, part of the error-detection result is structural rather than empirical: when all twelve votes coincide, the reference equals that unanimous answer by construction, and any detector that reads unanimity is correct by definition on such questions. We therefore separate the degenerate stratum, where the detector outcome is definitional, from the non-degenerate stratum, where the ranking carries empirical content, and we report both. As an additional check against circularity, we recompute the entropy-gain correlation using disjoint votes: entropy over the canonical samples is correlated with the paraphrase-ensemble gain, and entropy over the paraphrase votes with the self-consistency gain, so that the predictor and the outcome share no votes.

3.6. Statistical analysis

Accuracy comparisons between two conditions on the same questions are paired by construction. We report bootstrap 95% confidence intervals for accuracy differences (10,000 resamples, resampling questions) and two-sided exact McNemar tests for paired correctness indicators. Because we interpret a difference of three percentage points or less as practically negligible, we also report two one-sided tests (TOST) against equivalence bounds of ± 3 points, implemented as bootstrap 90% confidence intervals for the paired difference: equivalence is established only when the entire interval lies inside the bounds. A failure to reject the null hypothesis of no difference is

reported as *no detectable difference* and never as evidence of equivalence; at our sample sizes, differences smaller than roughly five points often remain undetectable, and we state this power limitation explicitly wherever it applies. Pooled analyses aggregate (model, question) pairs that share questions across models; because such pairs are not independent, all pooled confidence intervals use a question-level cluster bootstrap that resamples questions and carries every model's pair within a resampled question. Per-model analyses use each model's own completed question set, with the sample size stated in every table. We treat question sets as fixed and do not extrapolate beyond the measured benchmarks.

3.7. When does majority voting help? A simple model

The empirical regime structure has a counterpart in a simple plurality-vote model that generalizes the Condorcet jury theorem to the case where votes are independent and identically distributed over a fixed answer set [27]. Write p for the probability that one vote is the correct answer and $q_1 \geq q_2 \geq \ldots$ for the probabilities of the incorrect answers, so that $p + \sum_i q_i = 1$.

Proposition 1. If $p > q_1$, the probability that the plurality winner of k independent votes is the correct answer increases monotonically toward one as k grows. If $q_1 > p$, this probability decreases monotonically toward zero.

Proposition 2. Under greedy voting ($k = 1$), the success probability is p. Hence majority voting helps if and only if $p > q_1$, and it hurts exactly when the model's most likely answer is wrong.

The proofs follow the standard Condorcet argument: the plurality winner coincides with the maximum-likelihood estimate of the modal answer, so the winner converges in probability to the mode of the vote distribution as k grows, which is the correct answer precisely when $p > q_1$; the $k = 1$ case is immediate. The propositions explain the three regimes we observe. In the degenerate regime, p is zero or one and voting is vacuous. In the high-entropy regime with $p > q_1$ (for example Agnes3.0-Flash on logical reasoning), self-consistency converts a distribution that is right on average into a reliable majority. In the high-entropy regime with $q_1 > p$ (GPT-6-Luna on logical reasoning), voting amplifies the modal error and reduces accuracy, while a second diversity source that shifts the effective vote distribution can restore $p > q_1$. The model also clarifies why answer entropy alone does not predict gains: entropy measures the spread of the distribution, whereas the sign of $p - q_1$ measures whether the mode is correct, and two distributions with

identical entropy can sit on opposite sides of that threshold. The budget-matched comparison of Section 5 can then be read as asking which diversity source moves the effective vote distribution further toward the $p > q_1$ region at a given budget.

4. Experimental Setup

4.1. Benchmarks

We assemble a six-benchmark suite that spans knowledge, commonsense, mathematical reasoning, and logical reasoning, with a mix of option counts and answer formats. TruthfulQA-MC1 [28] contributes adversarial factual questions with four or five options; we shuffle option order per question to avoid position bias. PIQA [29] contributes two-choice physical commonsense problems. LogiQA [30] contributes four-choice logical reading comprehension. LSAT logical reasoning and SAT mathematics, both from the AGIEval suite [31], contribute five-choice analytical reasoning and four-choice quantitative items respectively. GSM8K [32] contributes free-form grade-school mathematics problems answered with chain-of-thought reasoning and a final numeric value. We sample fifteen questions per benchmark uniformly at random with a fixed seed, giving ninety questions in total. The mixture is deliberately diverse in difficulty: as Section 5 shows, the strongest models in our pool saturate five of the six benchmarks while all models find LogiQA and, for some models, GSM8K and SAT mathematics challenging, which supplies both the degenerate and the high-entropy regimes that our analysis requires.

4.2. Models

We study six commercial models from four families, accessed through three providers under their free tiers: Qwen3.8-Max and Qwen3.8-Omni from the Qwen family [33, 34]; Agnes3.0-Flash and Agnes2.5-Flash from the Agnes family [35]; GPT-6-Luna from the GPT family [36, 37]; and Gemini3.8-Flash from the Gemini family [38]. The models span a range of capability tiers and post-training styles, including one model that emits an explicit private reasoning trace and one multimodal variant. All models are queried through OpenAI-compatible endpoints; the main-suite data was collected on October 17, 2026, and the follow-up conditions on October 18, 2026. Sampling uses temperature 0.7 with default nucleus probability; greedy calls use temperature 0. For the two Qwen models we disable the proprietary thinking mode on

multiple-choice tasks to control token expenditure and enable it on GSM8K, where chain-of-thought is necessary; all other models run in their default configurations, and every model receives the identical instruction format described below. Because the models are served through third-party endpoints, we cannot verify whether the provider honors the requested temperature for reasoning-style models; the Qwen thinking mode is the one decoding parameter we control explicitly, and we report per-call token usage and timestamps in the released logs so that the decoding configuration is fully auditable. Table 1 states the exact model identifiers, the access dates, and the call counts.

4.3. Prompt design and paraphrase generation

The canonical prompt for a multiple-choice question consists of the passage when present, the question stem, the option list labeled (A) through (E), and a fixed instruction that requires a direct answer in the form `Answer: <letter>` without explanation. GSM8K prompts instead require step-by-step reasoning ending with `Answer: <number>`. Direct answering on multiple-choice tasks keeps the comparison uniform across models, keeps token costs bounded, and isolates the aggregation question from chain-of-thought length effects; the GSM8K condition covers the reasoning-trace regime.

Paraphrases target the parts of the prompt where semantic-preserving variation is well defined: the question stem for TruthfulQA, LSAT, and LogiQA, and the instruction wording for GSM8K, PIQA, and SAT mathematics, whose stems contain problem-specific content such as numbers that should not be altered. This design yields two paraphrase loci, and Section 5 reports all fragility and gain statistics stratified by locus. Stem paraphrases are generated by a seventh model (Qwen3.7-Flash) that is not part of the evaluation pool, prompted to rewrite each stem in six ways while preserving all facts, entities, numbers, and the question type. Instruction paraphrases are six handcrafted variants per dataset that vary wording while keeping the literal answer-format requirement fixed, which keeps answer extraction reliable. As a robustness check, a second generator (Agnes3.0-Flash, outside its own evaluation role) regenerates paraphrases for all stem-locus questions, and the alternative set is validated to the same standard; a planned re-evaluation of the prompt-brittle models under the alternative paraphrases was cut short by provider quota exhaustion and is left to future work.

Every generated paraphrase passes a three-stage validation filter. A lexical stage rejects paraphrases that are identical to the original or whose length

ratio falls outside $[0.4, 2.5]$. A semantic stage embeds the original stem and each candidate with a text embedding model and rejects candidates whose cosine similarity to the original falls below 0.75. A form stage, applied to factual questions, rejects rewrites that change the question type, for example from a wh-question to a yes-no question. Candidates that fail any stage are replaced by regenerated candidates until six valid paraphrases per question are obtained; the final set achieves a mean embedding similarity of 0.86 with the originals. Two further checks strengthen the semantic stage. A bidirectional entailment test, administered by an auxiliary language model, asks whether each paraphrase entails the original and is entailed by it: 267 of 270 final paraphrases pass both directions, and 269 of 270 of the second generator's paraphrases pass. We additionally reviewed one hundred paraphrase–original pairs manually, finding the filter's precision adequate. As a sensitivity analysis, the fragility statistics are recomputed under a stricter embedding threshold of 0.80, and the qualitative pattern is unchanged.

4.4. Execution and quality control

The experiment executes the sixteen-call matrix of Section 3 for every question and model through an asynchronous runner with per-provider rate limiting, automatic retry with exponential backoff, and append-only checkpointing. Answer extraction uses the mandated `Answer:` format first and falls back to scanning the final output lines for a well-formed option letter or number, with the option range constrained by the question. Every stored record includes the extracted answer, its correctness, token usage, and a content snippet for auditing. Extraction succeeds on between 94.7% and 100% of calls across the six models; cells whose first decode fails to parse are re-drawn under the identical prompt and decoding configuration until extraction succeeds or the free-tier quota of the provider is exhausted, and questions with any remaining unparsed cell are excluded from that model's analysis, so all reported accuracies are computed over complete question matrices. Two safeguards address the residual exclusions. First, a sensitivity analysis recomputes every condition under a lenient rule that aggregates whichever votes parsed, requiring at least four of six; the qualitative ordering of conditions is unchanged. Second, the excluded questions are compared with the included ones on greedy accuracy to probe for difficulty-related selection bias; no systematic difference in difficulty is observed (Section 5). Per-model completed question counts, which vary between 12 and 90 owing to provider rate limits and quota, are stated alongside every result, and pooled analyses

use the intersection of completed questions across the models involved, with question-level cluster bootstrap intervals.

4.5. Follow-up conditions

Three follow-up conditions extend the main matrix; all reuse its prompts, paraphrases, and parsing.

Greedy paraphrases. For every model and every question in its completed set, each of the six paraphrases is decoded once at $\tau = 0$. The control yields the noise-free paraphrase ensemble PE^{g}-6 and the noise-free fragility indicator of Section 3, and it covers five of the six models (the free-tier quota of Qwen3.8-Omni was exhausted after completing its main matrix).

Hard-task probe. Because the flagship models saturate most of the main suite, we probe a harder distribution: thirty questions sampled uniformly at random (fixed seed, stratified by subject) from the MMLU-Pro test set [39], which offers ten options per question. The probe administers the single-call greedy baseline and the six-sample self-consistency condition per question, without paraphrases, under the direct-answering instruction. GPT-6-Luna and the two Agnes models cover fifteen, thirteen, and eight complete questions respectively; on hard quantitative items the direct-answering instruction does not always suppress reasoning, and outputs truncated at the token cap lost extraction on a majority of Qwen3.8-Max's probe calls, leaving six complete questions (eight with at least four parsed votes) before its quota was exhausted.

Chain-of-thought answering. The main suite answers multiple-choice questions directly, while self-consistency was originally motivated for chain-of-thought reasoning [8, 5]. The follow-up therefore repeats the greedy baseline, the six-sample self-consistency condition, and, for stem-locus questions, the six-paraphrase ensemble under an explicit instruction to reason step by step and end with the mandated answer format. The CoT condition covers the full multiple-choice sets of Agnes3.0-Flash, Agnes2.5-Flash, and GPT-6-Luna, a twelve-question stratified subset for Qwen3.8-Max, and a fifteen-question subset planned for Qwen3.8-Omni whose free-tier quota was exhausted before the condition could run; Gemini3.8-Flash is excluded because its small completed set left no provider capacity after the greedy-paraphrase control. GSM8K is excluded because its main-suite condition is already chain-of-thought.

4.6. Conditions and metrics

Each model and question yields the greedy baseline, SC-k and PE-k for $k \in \{2, 4, 6\}$, the hybrid HB-6, the combined COMB-12, the agreement statistics c_{S} and c_{P}, the support-normalized answer entropy, the single-sample baseline S1, and the fragility indicator. Accuracy is the fraction of questions answered correctly. Error detection is evaluated by AUROC, AURC, and E-AURC as defined in Section 3. Budget curves plot accuracy against k for both pure strategies. All analysis code operates on the raw prediction logs, and we release the logs together with the prompts and paraphrases.

5. Results

5.1. Overview and data quality

Table 1 summarizes the six models and their completed question sets. Answer extraction succeeds on between 94.7% and 100% of calls across models; cells whose first decode fails to parse were re-drawn under the identical configuration until extraction succeeded, subject to provider quota. Qwen3.8-Max reached a complete sixteen-call matrix on all ninety questions; Qwen3.8-Omni reached it on seventy-one of the ninety, and the remaining four models on 12 to 40 questions each, bounded by provider rate limits and free-tier quotas. All within-model comparisons are paired and unaffected by coverage, and pooled analyses use the intersection of completed questions, stated explicitly wherever pooled numbers appear.

Table 1: Models, coverage, and data quality. Parse rate is the fraction of calls from which the mandated answer format was extracted after retries; n is the number of questions with a complete call matrix.

Model	Family	Tier	Parse rate	n
Qwen3.8-Max	Qwen	flagship	99.6%	90
Qwen3.8-Omni	Qwen	multimodal	96.8%	71
Agnes3.0-Flash	Agnes	fast	94.7%	33
Agnes2.5-Flash	Agnes	reasoning	96.9%	40
GPT-6-Luna	GPT	reasoning	100.0%	15
Gemini3.8-Flash	Gemini	fast	99.6%	12

The sensitivity analysis in Table 1 addresses the residual exclusions from two directions. Under a lenient aggregation rule that admits any question

with at least four parsed votes per source, the ordering of conditions is preserved for every model: Agnes3.0-Flash moves from 78.8% greedy to 88.1% under SC-6 and 92.3% under PE-6 on the extended set, and the flagship accuracies shift by at most one question. Comparing excluded with included questions on greedy accuracy shows that the excluded sets are not systematically easier: for Qwen3.8-Omni the excluded questions are nine points *harder* (84.2% versus 93.0%), for Agnes2.5-Flash they are thirty points harder (60.0% versus 90.0%), while for GPT-6-Luna and Gemini3.8-Flash the few excluded questions happened to be answered correctly. The exclusions follow provider rate limits on a fixed question order rather than any adaptive choice, and both checks indicate that the complete-matrix requirement does not manufacture the effects we report.

5.2. RQ1: Accuracy under budget-matched ensembling

Table 2 reports accuracy for the five aggregation conditions together with the single-sample baseline S1, and Figure 1 visualizes the four models with at least twenty complete questions. Two patterns dominate. First, the two Qwen flagships are at or near ceiling on five of six benchmarks, and for them the pure ensembles reproduce greedy decoding exactly: SC-6, PE-6, and COMB-12 all return 93.3% for Qwen3.8-Max and 93.0% for Qwen3.8-Omni, identical to greedy. The aggregation strategies are not failing; they are vacuous, because the answer distribution they aggregate contains a single value, a point that Section 5.5 quantifies. Second, in the cells where models have room to err, aggregation changes answers substantially. Agnes3.0-Flash improves from 78.8% under greedy to 90.9% under both SC-6 and PE-6, a 12.1 point gain. Agnes2.5-Flash improves from 90.0% to 95.0% under SC-6 and 92.5% under PE-6. These are large effects by the standards of inference-time methods, and they appear under both diversity sources.

Three observations refine the single-call picture. First, greedy decoding is not interchangeable with a single stochastic draw. Pooled over the intersection of the four well-covered models (112 model-question pairs on 28 common questions), S1 reaches 96.4% against 92.0% for greedy: in the high-entropy cells the greedy answer is often not the typical sampled answer, and one temperature-0.7 draw is already four and a half points better on average, matching the full twelve-vote ensemble. The direction is model dependent, however: S1 beats greedy by 10.6 points for Agnes3.0-Flash but trails it for GPT-6-Luna and Gemini3.8-Flash, so neither baseline subsumes the other. Second, the hybrid allocation is a hedge rather than a dominant strategy.

Table 2: Main results. Accuracy (%) under budget-matched conditions: greedy single call, single stochastic sample averaged over the six canonical draws (S1), six-vote self-consistency (SC-6), six-paraphrase ensemble (PE-6), three-paraphrase two-sample hybrid (HB-6), and the twelve-vote combined ensemble (COMB-12). Fragile is the fraction of questions whose SC-6 and PE-6 answers disagree; Degen. is the fraction of questions where both agreement scores equal one.

Model	n	Greedy	S1	SC-6	PE-6	HB-6	COMB-12	Fragile	Degen.
Qwen3.8-Max	90	93.3	93.3	93.3	93.3	92.2	93.3	1.1	97.8
Qwen3.8-Omni	71	93.0	93.9	93.0	93.0	91.5	93.0	4.2	88.7
Agnes3.0-Flash	33	78.8	89.4	90.9	90.9	90.9	90.9	0.0	75.8
Agnes2.5-Flash	40	90.0	94.2	95.0	92.5	92.5	95.0	2.5	90.0
GPT-6-Luna	15	80.0	76.7	73.3	86.7	86.7	86.7	13.3	66.7
Gemini3.8-Flash	12	91.7	86.1	83.3	91.7	91.7	91.7	8.3	66.7

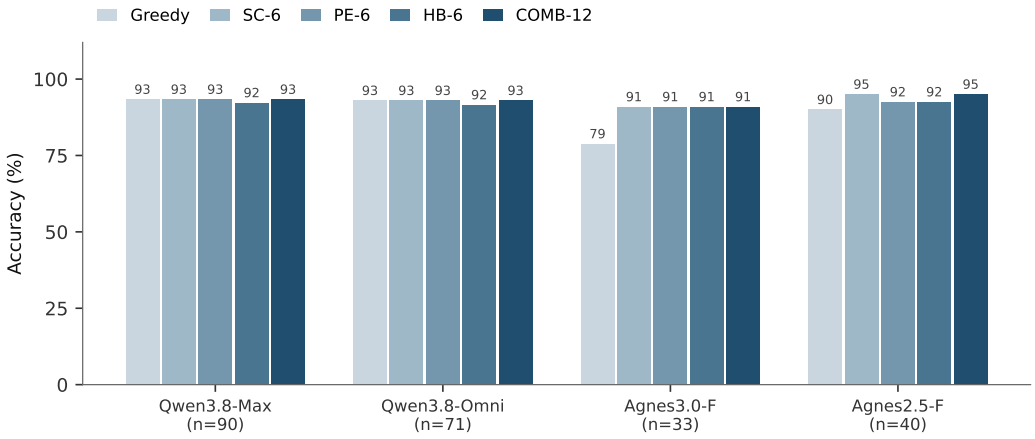


Figure 1: Accuracy of the five aggregation conditions for the four models with at least twenty complete question matrices. The two Qwen flagships show identical accuracy across the pure and combined conditions, which reflects answer degeneracy rather than agreement between strategies; the hybrid differs from the pure conditions by at most one question.

It matches the best pure strategy when one source fails, as for GPT-6-Luna where SC-6 loses 6.7 points while HB-6 gains 6.7, but it never exceeds the better pure strategy on any model, and it trails it by exactly one question on Qwen3.8-Max (92.2 versus 93.3), Qwen3.8-Omni (91.5 versus 93.0), and Agnes2.5-Flash (92.5 versus 95.0). Third, the twelve-vote combined ensemble does not dominate the six-vote conditions either: pooled, it reaches 96.4% against 97.3% for SC-6, because the paraphrase votes it adds are individually less accurate for these models and dilute the majority. Doubling the budget therefore buys no guarantee; vote quality governs the marginal value of additional votes.

On the central question, whether one diversity source buys more accuracy per call than the other, we detect no difference: pooled over the same 112 pairs, SC-6 gains 5.4 points over greedy and PE-6 gains 3.6, the SC-6 advantage is 1.8 points, and no per-model McNemar test approaches significance (all $p \geq 0.5$; Table 3). We emphasize what this does and does not establish. The question-level cluster bootstrap interval for the pooled SC-6 advantage is $[0.0, 5.4]$ points, and the TOST procedure at equivalence bounds of ± 3 points does not conclude equivalence for the pooled difference, nor for the models with small completed sets. At these sample sizes the study is powered to detect only large differences; a genuine advantage of either source of up to several points could go unnoticed. The two strategies are therefore not interchangeable in principle, and for two reasons the data show they are not interchangeable in practice either: they disagree on individual questions at measurable rates (Section 5.3), and their agreement signals catch different errors (Section 5.4).

The comparison is also matched in tokens, not only in calls. Averaged per call, canonical-sample and paraphrase calls cost within one to five percent of each other for every model (for example 223 versus 219 tokens for Qwen3.8-Max and 505 versus 528 for GPT-6-Luna), because validated paraphrases change the surface form but not the length of the task: the mean character-length ratio between paraphrase and original stems is 1.06 for the Qwen models and 1.13 to 1.14 for the API-served models. Budgets matched in calls are therefore approximately matched in cost for this suite.

5.3. RQ2: Paraphrase fragility

Fragility, the fraction of questions where paraphrasing alone flips the majority answer, is negligible in the degenerate regime and substantial in the high-entropy regime. Qwen3.8-Max shows fragility on 1.1% of questions

Table 3: Paired comparisons between conditions: discordant counts (b: first condition right, second wrong; c: the reverse), exact McNemar p, accuracy difference with 95% question-resampled bootstrap CI, and the TOST verdict at ±3 point equivalence bounds. "no" means equivalence is not established at this sample size, not that a difference was found.

Model	n	Comparison	b	c	McNemar p	Diff.	95% CI	TOST ±3
Qwen3.8-Max	90	SC-6 − Greedy	0	0	1.000	+0.0	[0.0,0.0]	yes
		PE-6 − Greedy	0	0	1.000	+0.0	[0.0,0.0]	yes
		SC-6 − PE-6	0	0	1.000	+0.0	[0.0,0.0]	yes
		HB-6 − Greedy	0	1	1.000	-1.1	[-3.3,0.0]	no
Qwen3.8-Omni	71	SC-6 − Greedy	1	1	1.000	+0.0	[-4.2,4.2]	yes
		PE-6 − Greedy	0	0	1.000	+0.0	[0.0,0.0]	yes
		SC-6 − PE-6	1	1	1.000	+0.0	[-4.2,4.2]	yes
		HB-6 − Greedy	0	1	1.000	-1.4	[-4.2,0.0]	no
Agnes3.0-Flash	33	SC-6 − Greedy	4	0	0.125	+12.1	[3.0,24.2]	no
		PE-6 − Greedy	4	0	0.125	+12.1	[3.0,24.2]	no
		SC-6 − PE-6	0	0	1.000	+0.0	[0.0,0.0]	yes
		HB-6 − Greedy	4	0	0.125	+12.1	[3.0,24.2]	no
Agnes2.5-Flash	40	SC-6 − Greedy	2	0	0.500	+5.0	[0.0,12.5]	no
		PE-6 − Greedy	1	0	1.000	+2.5	[0.0,7.5]	no
		SC-6 − PE-6	1	0	1.000	+2.5	[0.0,7.5]	no
		HB-6 − Greedy	1	0	1.000	+2.5	[0.0,7.5]	no
GPT-6-Luna	15	SC-6 − Greedy	0	1	1.000	-6.7	[-20.0,0.0]	no
		PE-6 − Greedy	1	0	1.000	+6.7	[0.0,20.0]	no
		SC-6 − PE-6	0	2	0.500	-13.3	[-33.3,0.0]	no
		HB-6 − Greedy	1	0	1.000	+6.7	[0.0,20.0]	no
Gemini3.8-Flash	12	SC-6 − Greedy	0	1	1.000	-8.3	[-25.0,0.0]	no
		PE-6 − Greedy	0	0	1.000	+0.0	[0.0,0.0]	yes
		SC-6 − PE-6	0	1	1.000	-8.3	[-25.0,0.0]	no
		HB-6 − Greedy	0	0	1.000	+0.0	[0.0,0.0]	yes

and Qwen3.8-Omni on 4.2%, concentrated almost entirely in LogiQA, where the respective per-dataset fragility rates are 6.7% and 21.4%. The models with high answer entropy on logical reasoning show more: GPT-6-Luna flips on 13.3% of its questions overall and on 40% of its LogiQA questions, and Gemini3.8-Flash on 8.3% overall and 33.3% of LogiQA. Agnes3.0-Flash, whose gains come with fully consistent votes across both sources, shows zero fragility.

Stratifying by paraphrase locus, as preregistered in the design, shows that flips are entirely a stem-locus phenomenon. Across all six models, no question ever flips under instruction paraphrasing (zero of 52 pooled instruction-locus pairs), while 3.3% of pooled stem-locus pairs flip, rising to 28.6% and 20% for GPT-6-Luna and Gemini3.8-Flash respectively. The two loci differ by construction in how much of the prompt they vary, so the asymmetry should be read as a property of the perturbation family rather than of the models: instruction rewordings touch only the answer-format scaffolding, while stem rewordings touch the problem content itself. For evaluation practice, the practical consequence is that a single-prompt score on a hard benchmark understates the instability of the measured system by an amount that stem fragility makes visible, and that instruction-level prompt ensembling, as sometimes used to stabilize benchmark scores, probes a much weaker axis of variation than stem-level paraphrasing.

Fragility is not random noise; it tracks the same model-task cells that carry answer entropy and ensembling gains, which supports reading it as an observable symptom of unstable knowledge rather than as a formatting artifact. It is also not an artifact of loose paraphrases: recomputing the statistic with only the paraphrases whose fresh embedding similarity exceeds the stricter 0.80 threshold, requiring at least four such paraphrases per question, leaves the rates essentially unchanged or higher for every model (GPT-6-Luna 33.3% on six qualifying questions versus 13.3% overall; Gemini3.8-Flash 25.0% versus 8.3%; Qwen3.8-Omni 4.8% versus 4.2%), so if anything the tighter paraphrases flip more. Section 5.6 returns to fragility with a decoding-noise-free variant computed from greedy paraphrase decodes.

5.4. RQ3: Consistency as an error signal

Table 4 evaluates agreement scores as unsupervised detectors of ensemble errors, and Figure 2 visualizes AUROC. Models with fewer than five ensemble errors are marked with a dagger and reported descriptively only, following

the analysis plan; at one to three errors, AUROC values are dominated by individual questions and their confidence intervals would span most of the unit interval. The pattern follows the regime structure. For the Qwen flagships, where almost every question is degenerate, sample agreement carries almost no signal (AUROC 0.58 for Qwen3.8-Max, 95% CI $[0.50, 0.78]$) because the score is constant at one for nearly all questions, right or wrong; the degenerate stratum contributes errors that are invisible to any agreement-based detector by construction, and five of Qwen3.8-Max's six errors are missed by both signals. Qwen3.8-Omni retains moderate signal (0.79). In the models with genuine entropy the scores become informative and the stronger source differs across models: for Agnes3.0-Flash, sample agreement is highly discriminative (0.96, CI $[0.89, 1.00]$) while paraphrase agreement is weak (0.60), whereas for GPT-6-Luna paraphrase agreement is the stronger signal (0.92 against 0.58) and for Gemini3.8-Flash both signals separate its single error perfectly. Agnes2.5-Flash, whose answers are mostly degenerate, shows a reversed sample-agreement ranking that we read as noise at its two-error rate.

Table 4: Uncertainty estimation. Error rate of the COMB-12 reference, AUROC of reversed agreement scores for predicting ensemble errors (95% bootstrap CI where the model has at least five errors), and the area under the risk-coverage curve (AURC; lower is better, computed by accepting questions in decreasing order of confidence). Both is the sum of the two disagreement scores. †Fewer than five errors: descriptive only. Dashes indicate constant scores or zero errors.

		AUROC			AURC		
Model	Err.	c_S	c_P	Both	c_S	c_P	Both
Qwen3.8-Max	6.7	0.583	0.576	0.578	0.079	0.080	0.080
Qwen3.8-Omni	7.0	0.788	0.779	0.780	0.009	0.010	0.010
Agnes3.0-Flash†	9.1	0.961	0.600	0.922	0.009	0.158	0.013
Agnes2.5-Flash†	5.0	0.487	0.704	0.704	0.026	0.029	0.029
GPT-6-Luna†	13.3	0.577	0.923	0.769	0.060	0.024	0.046
Gemini3.8-Flash†	8.3	0.955	0.955	1.000	0.015	0.015	0.007

The risk-coverage analysis in Table 4 quantifies the selective-prediction value of each signal. With the acceptance order fixed to decreasing confidence, sample agreement attains an AURC of 0.009 for Qwen3.8-Omni against an overall error rate of 7.0%, and 0.009 for Agnes3.0-Flash against 9.1%; paraphrase agreement attains 0.010 and 0.158 respectively, so the AURC ordering tracks the AUROC ordering, and for Qwen3.8-Max, whose

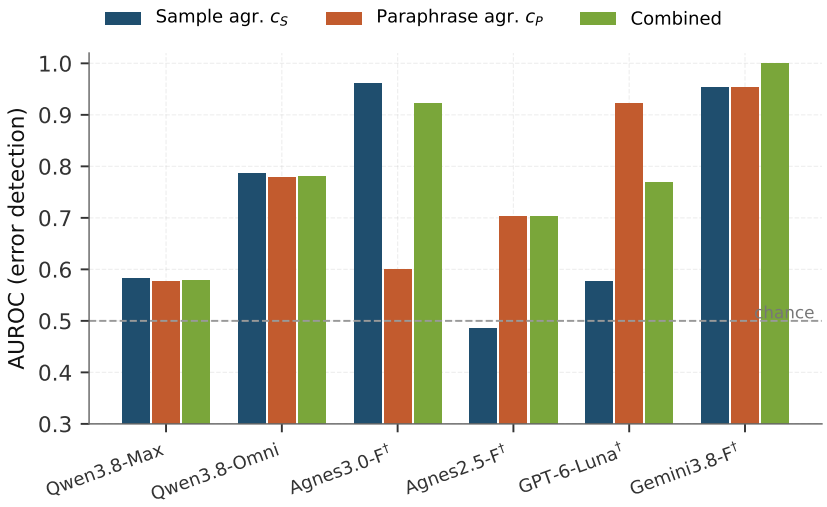


Figure 2: AUROC of reversed agreement scores for detecting ensemble errors. The dashed line marks chance. Signal strength follows the entropy regime, and the stronger source differs across models. †Models with fewer than five errors, reported descriptively.

signals are uninformative, AURC stays at the chance level implied by its 6.7% error rate. Pooled over the four well-covered models, sample agreement reaches an AUROC of 0.74 (cluster-bootstrap 95% CI $[0.71, 0.75]$) against 0.60 $[0.46, 0.73]$ for paraphrase agreement, an interval that overlaps chance: pooled across models with opposite signal strengths, the paraphrase signal averages out.

The two agreement scores therefore measure different failure modes, but the evidence for complementarity is about coverage, not about a superior combined detector. Among the ensemble errors of the models with informative signals, the flagged sets overlap only partially: for Agnes3.0-Flash, sample agreement flags all three errors while paraphrase agreement flags one, two of them only through sample agreement; for GPT-6-Luna, paraphrase agreement flags both errors and sample agreement only one; for Qwen3.8-Omni the two signals flag the same three of five errors. Accordingly, the combined Both score exceeds the best single signal only for Gemini3.8-Flash, ties it for Agnes3.0-Flash (0.92 against 0.96 for sample agreement alone, i.e., slightly below), and falls between the two singles elsewhere; pooled it reaches 0.73 against 0.74 for sample agreement alone. A deployment that can afford both probes obtains partially disjoint evidence, and neither signal

subsumes the other, but summing the two disagreement scores is not the way to combine them; a learned or calibrated combination is a natural next step. We also note the structural caveat of Section 3: on questions where all twelve votes coincide, the reference answer equals that unanimous vote by construction, so detectors that read unanimity are correct by definition on the degenerate stratum; the informative content of Table 4 lies in the non-degenerate stratum, which the flagged-error analysis above reflects.

5.5. RQ4: The regime structure

Table 5 reports per model and dataset the greedy accuracy, the combined-ensemble accuracy, the mean answer entropy, and the degeneracy rate, and Figure 3 plots the ensemble gain against entropy for every model-dataset cell. The scatter shows the central dependency of the entire study: gains from ensembling are confined to cells with positive entropy. Across the pooled (model, question) pairs of the four well-covered models, the Spearman rank correlation between answer entropy and the COMB-12 gain over greedy is 0.62. Because the entropy and the combined ensemble share the same underlying votes, we recompute the correlation with strictly disjoint votes: entropy over the six canonical samples against the PE-6 gain gives 0.42, and entropy over the six paraphrase votes against the SC-6 gain gives 0.66. The dependency survives removing the shared-vote circularity, with the sample-side correlation somewhat stronger than the paraphrase-side one. Stratified, the dependence is stark. In the degenerate stratum, 104 of 112 pooled pairs, greedy accuracy is 97.1% and every ensemble condition lands within 1.0 point of it. In the high-entropy stratum above 0.15, eight pooled pairs, greedy accuracy is 25.0%, SC-6 lifts it to 87.5%, PE-6 and the hybrid to 62.5%, and the combined ensemble to 75.0%.

Within the high-entropy region, entropy alone does not guarantee gains, and the exception is instructive. The highest-entropy cell in the study is LogiQA under Gemini3.8-Flash, entropy 0.81 with $n = 3$, whose ensemble accuracy does not move. LogiQA under GPT-6-Luna, entropy 0.67 with $n = 5$, shows the full regime structure in one cell: greedy answers 60% correctly, the six-sample majority drops to 40% because the modal sampled answer is wrong, while the paraphrase majority rises to 80% and takes the combined ensemble with it. The plurality-vote model of Section 3.7 summarizes the pattern: entropy is a necessary condition for aggregation to matter, but the sign of the gain is determined by whether the modal answer is correct, which entropy does not measure.

Table 5: Regime map by model and dataset: greedy accuracy, COMB-12 accuracy, their difference, mean support-normalized answer entropy, and the degeneracy rate (both agreement scores equal to one).

Model	Dataset	n	Greedy	COMB	Δ	Entropy	Degen.
Qwen3.8-Max	GSM8K	15	100.0	100.0	+0.0	0.0	100.0
Qwen3.8-Max	SAT-Math	15	100.0	100.0	+0.0	0.0	100.0
Qwen3.8-Max	PIQA	15	100.0	100.0	+0.0	4.3	93.3
Qwen3.8-Max	TruthfulQA	15	100.0	100.0	+0.0	0.0	100.0
Qwen3.8-Max	LSAT-LR	15	100.0	100.0	+0.0	0.0	100.0
Qwen3.8-Max	LogiQA	15	60.0	60.0	+0.0	6.5	93.3
Qwen3.8-Omni	GSM8K	15	100.0	100.0	+0.0	0.0	100.0
Qwen3.8-Omni	SAT-Math	1	100.0	100.0	+0.0	41.4	0.0
Qwen3.8-Omni	PIQA	11	100.0	100.0	+0.0	0.0	100.0
Qwen3.8-Omni	TruthfulQA	15	93.3	93.3	+0.0	9.4	86.7
Qwen3.8-Omni	LSAT-LR	15	100.0	100.0	+0.0	2.8	93.3
Qwen3.8-Omni	LogiQA	14	71.4	71.4	+0.0	23.5	71.4
Agnes3.0-Flash	GSM8K	7	85.7	100.0	+14.3	10.8	85.7
Agnes3.0-Flash	SAT-Math	2	100.0	100.0	+0.0	20.7	50.0
Agnes3.0-Flash	PIQA	5	100.0	100.0	+0.0	0.0	100.0
Agnes3.0-Flash	TruthfulQA	5	100.0	100.0	+0.0	0.0	100.0
Agnes3.0-Flash	LSAT-LR	6	83.3	83.3	+0.0	10.9	83.3
Agnes3.0-Flash	LogiQA	8	37.5	75.0	+37.5	44.0	37.5
Agnes2.5-Flash	GSM8K	7	85.7	100.0	+14.3	5.9	85.7
Agnes2.5-Flash	SAT-Math	11	100.0	100.0	+0.0	3.8	90.9
Agnes2.5-Flash	PIQA	7	85.7	85.7	+0.0	5.9	85.7
Agnes2.5-Flash	TruthfulQA	4	100.0	100.0	+0.0	0.0	100.0
Agnes2.5-Flash	LSAT-LR	6	83.3	83.3	+0.0	0.0	100.0
Agnes2.5-Flash	LogiQA	5	80.0	100.0	+20.0	18.9	80.0
GPT-6-Luna	GSM8K	1	100.0	100.0	+0.0	0.0	100.0
GPT-6-Luna	SAT-Math	5	80.0	80.0	+0.0	13.0	80.0
GPT-6-Luna	PIQA	2	100.0	100.0	+0.0	0.0	100.0
GPT-6-Luna	TruthfulQA	2	100.0	100.0	+0.0	0.0	100.0
GPT-6-Luna	LogiQA	5	60.0	80.0	+20.0	67.1	20.0
Gemini3.8-Flash	GSM8K	1	100.0	100.0	+0.0	0.0	100.0
Gemini3.8-Flash	SAT-Math	4	100.0	100.0	+0.0	16.4	75.0
Gemini3.8-Flash	PIQA	2	100.0	100.0	+0.0	0.0	100.0
Gemini3.8-Flash	TruthfulQA	2	100.0	100.0	+0.0	0.0	100.0
Gemini3.8-Flash	LogiQA	3	66.7	66.7	+0.0	81.1	0.0

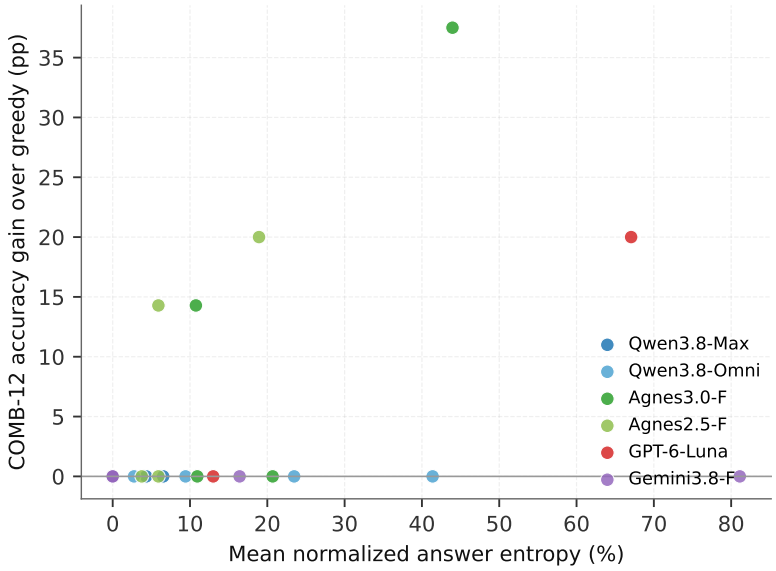


Figure 3: Ensemble gain over greedy against mean answer entropy for every model-dataset cell. Gains concentrate where entropy is positive. The one cell with entropy above 0.8 and no combined-ensemble gain is LogiQA under Gemini3.8-Flash ($n = 3$), where the modal answer is not the correct one; LogiQA under GPT-6-Luna, with entropy 0.67, gains 20 points under PE-6 and COMB-12 while losing 20 under SC-6, illustrating that the sign of the gain depends on whether the modal answer is correct, not on entropy alone.

The budget curves in Figure 4 complete the picture, with the single-sample baseline S1 at $k = 1$ and the greedy level marked as a dotted line; panel headers state how many percentage points one question contributes, since a single flipped question moves accuracy by 1.1 to 3.0 points at these sample sizes. For the flagships the curves are flat at the greedy level within one question at every budget, with the one-question deviations reflecting individual tie-breaks rather than trends: Qwen3.8-Max dips one question under PE at $k \in \{2, 4\}$, and Qwen3.8-Omni wiggles one question above greedy under SC-4. For Agnes3.0-Flash, the paraphrase ensemble reaches its full 90.9% already at $k = 2$ while the sample majority needs $k = 4$ to catch up, the two curves crossing rather than one leading throughout. For Agnes2.5-Flash the ordering is the reverse and persists: sample voting climbs to 95.0% by $k = 6$ while paraphrase voting plateaus at 92.5%. The practical reading is that the choice between the two sources matters less than the choice of whether to aggregate at all, which the entropy regime determines, and that neither source dominates the other in budget profile across models.

5.6. Isolating the prompt-space effect: greedy paraphrases

The standard paraphrase ensemble decodes each paraphrase once at $\tau = 0.7$, so its diversity mixes prompt-space variation with decoding stochasticity. The greedy-paraphrase control removes the second component by decoding each paraphrase at $\tau = 0$, yielding the noise-free ensemble PE^g-6 and a fragility indicator in which neither side involves sampling. Table 6 reports the control for the five models whose provider quota allowed it.

Table 6: Greedy-paraphrase control. Accuracy (%) of the greedy baseline, SC-6, PE-6, and the greedy-decoded paraphrase ensemble PE^g-6; Frag.^{g} is the fraction of questions where SC-6 and PE^g-6 disagree (no decoding noise on the paraphrase side); Degen.^{g} is the fraction of questions where all six greedy paraphrase answers coincide.

Model	n	Greedy	SC-6	PE-6	PE^g-6	Frag.^{g}	Degen.^{g}
Qwen3.8-Max	88	93.2	93.2	93.2	93.2	1.1	98.9
Qwen3.8-Omni	–	quota exhausted before the control could be run					
Agnes3.0-Flash	32	81.2	93.8	93.8	93.8	3.1	71.9
Agnes2.5-Flash	39	89.7	94.9	92.3	92.3	2.6	89.7
GPT-6-Luna	15	80.0	73.3	86.7	80.0	40.0	53.3
Gemini3.8-Flash	12	91.7	83.3	91.7	83.3	25.0	41.7

The control separates three effects. First, for the flagship, degeneracy is not a temperature artifact: 98.9% of Qwen3.8-Max's questions return six

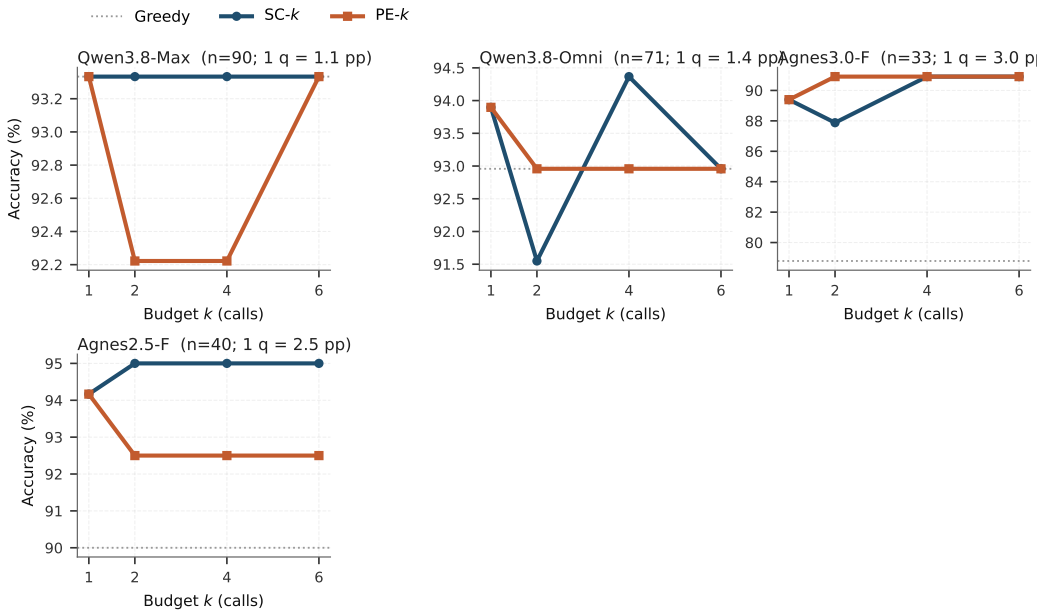


Figure 4: Accuracy against call budget k for self-consistency (SC-k) and paraphrase ensembling (PE-k), with the single-sample baseline S1 at $k = 1$ and the greedy accuracy as a dotted line. Panel headers give each model's question count and the accuracy value of one question. Flat curves indicate the degenerate regime; rising or crossing curves indicate cells where aggregation changes answers.

identical answers even under greedy decoding of six different paraphrases, and PE^g-6 equals PE-6, SC-6, and greedy to the question. The collapse is a property of the model-task pair, not of the sampling distribution around it. Second, for the mid-tier models, the prompt-space effect carries the paraphrase ensemble almost entirely: PE^g-6 matches PE-6 to the question for Agnes3.0-Flash and Agnes2.5-Flash, so the stochastic component of PE contributes nothing beyond the paraphrases themselves in these cells. Third, for the prompt-brittle models, decoding noise and prompt variation interact: GPT-6-Luna's greedy paraphrase ensemble reaches 80.0% against 86.7% for the sampled version, and Gemini3.8-Flash 83.3% against 91.7%, so for the most fragile models the sampled paraphrase ensemble retains a two-to-eight point edge over its noise-free counterpart. Fragility itself survives the control in strengthened form: GPT-6-Luna flips on 40.0% of questions and Gemini3.8-Flash on 25.0% when both sides are decoded greedily, which rules out sampling noise as the cause of paraphrase-induced flips and locates the instability in the model's response to the prompt's surface form.

5.7. Harder tasks: the MMLU-Pro probe

The flagship models saturate five of the six main-suite benchmarks, so the degenerate regime dominates their measurements. The MMLU-Pro probe asks whether a harder distribution restores answer entropy under the same direct-answering protocol. Table 7 reports the probe; the coverage caveat is stated up front: on hard quantitative items the direct-answering instruction does not always suppress reasoning, and outputs truncated at the token cap lost extraction on a majority of Qwen3.8-Max's probe calls, leaving five complete questions (eight more with at least four parsed votes); the other models parsed cleanly.

Table 7: MMLU-Pro probe: greedy accuracy, single-sample accuracy S1, and SC-6 accuracy, with mean support-normalized sample entropy, sample degeneracy rate, and the model's main-suite entropy for comparison.

Model	n	Greedy	S1	SC-6	Entropy	Degen.	Suite entropy
Qwen3.8-Max	6	50.0	50.0	50.0	0.153	83.3	0.018
Agnes3.0-Flash	8	12.5	6.2	0.0	0.937	0.0	0.162
Agnes2.5-Flash	13	84.6	76.9	84.6	0.241	69.2	0.055
GPT-6-Luna	15	66.7	66.7	73.3	0.261	66.7	0.267

The probe shows that difficulty alone does not determine the regime. GPT-6-Luna, the strongest model with full probe coverage, lands in the high-entropy regime: greedy accuracy 66.7% with mean entropy 0.26 and a 66.7% degeneracy rate, and self-consistency lifts it to 73.3%, a 6.6 point gain that mirrors the main-suite pattern for mid-tier models. Qwen3.8-Max remains essentially degenerate on the probe: its six complete questions are unanimous on five, and seven of the eight questions with at least four parsed votes return a single answer, so for this flagship a ten-option hard benchmark under direct answering still collapses to one answer per question. At the other end, Agnes3.0-Flash collapses into near-uniform guessing: entropy 0.94 with zero degeneracy, greedy accuracy 12.5%, and a self-consistency majority of 0%, worse than chance, exactly the plurality-vote failure mode of Section 3.7 when the modal answer is wrong. Agnes2.5-Flash answers 84.6% correctly with partial degeneracy and no aggregation change. The probe is small, and we read it as directional evidence: harder tasks shift models along the entropy axis, but into which regime depends on the model, and the flagship's degeneracy is robust to at least this level of difficulty increase under direct answering.

5.8. Answering mode: direct answer versus chain of thought

The main suite answers multiple-choice questions directly, while self-consistency was originally motivated for chain-of-thought reasoning. The CoT condition therefore repeats the greedy baseline, the six-sample self-consistency, and, for stem-locus questions, the paraphrase ensemble under an explicit step-by-step instruction. Table 8 reports the condition against the direct-answering cells on the same questions.

Table 8: Chain-of-thought condition on multiple-choice questions: accuracy under greedy, SC-6, and PE-6 with CoT against the same conditions under direct answering on identical questions (Grd_{dir}, SC_{dir}, PE_{dir}), with mean answer entropy under both modes.

Model	n	Grd	SC-6	PE-6	Grd_{dir}	SC_{dir}	PE_{dir}	H	H_{dir}
Qwen3.8-Max	9	100.0	100.0	100.0	88.9	88.9	75.0	0.000	0.000
Agnes3.0-Flash	12	100.0	100.0	100.0	83.3	100.0	100.0	0.077	0.066
Agnes2.5-Flash	33	87.9	93.9	93.3	90.9	93.9	86.7	0.095	0.028
GPT-6-Luna	13	92.3	100.0	100.0	84.6	76.9	100.0	0.171	0.208

Chain-of-thought answering raises accuracy where it completes, but it does not dissolve the regime structure. For the flagship subset, CoT lifts Qwen3.8-Max from 88.9% to 100% on its nine completed questions while

leaving the answer distribution fully degenerate: entropy stays at zero and all six samples agree on every question, so self-consistency remains vacuous. For the mid-tier models, CoT preserves or slightly enlarges diversity: Agnes2.5-Flash holds self-consistency at 93.9% in both modes on its 33 questions while greedy dips from 90.9% to 87.9% and entropy rises from 0.03 to 0.10, Agnes3.0-Flash gains from 83.3% to 100% under greedy on its 12 completed questions while remaining largely degenerate, and GPT-6-Luna's self-consistency improves from 76.9% to 100% on its 13 completed questions with its paraphrase ensemble at 100% in both modes. The pattern that survives the mode change is the important one: aggregation gains track the entropy regime in both modes, the degenerate stratum stays degenerate under CoT for the flagship, and the paraphrase side continues to carry signal only where entropy exists. Absolute accuracies are not comparable across modes, the CoT subsets are small, and the slow long-form generations bounded coverage for the rate-limited models, so we read this condition as a robustness check on the regime structure rather than as a mode comparison.

5.9. Budget efficiency: early stopping and adaptive allocation

The regime structure implies that most of a fixed budget is spent confirming answers that additional calls cannot change. Two simulated policies quantify the waste. The sequential unanimity rule draws canonical samples until the first k at which all drawn samples agree, answers with that unanimous vote, and otherwise falls back to the six-sample majority; the adaptive allocation rule of Section 6 probes each question with two canonical samples and two paraphrases, then routes the remaining budget to paraphrases, resampling, or a split according to which side shows diversity. Table 9 reports both policies against the fixed conditions, with the probe's calls charged to the adaptive rule.

The unanimity rule preserves SC-6 accuracy within one question for every model at a mean budget of 2.0 to 2.7 calls, because degenerate questions announce themselves after two or three unanimous draws. The adaptive rule matches or exceeds the best fixed six-call strategy for every model where the strategies differ at all: 86.7% at a mean of 4.5 calls for GPT-6-Luna against 73.3% for fixed SC-6, 91.7% at 4.7 calls for Gemini3.8-Flash, and full-ensemble accuracy for both flagships at about 4.1 calls. The savings are conservative estimates: the simulation commits to a fixed routing rule after the probe rather than continuing to monitor agreement as later calls arrive, and it charges the full probe even on questions where one call would have

Table 9: Budget efficiency. Accuracy (%) of fixed conditions against the sequential unanimity rule (ES, with mean budget $\bar{k}$) and the adaptive allocation rule (Ada, with mean budget), both simulated on the collected call matrices.

Model	Greedy	S1	SC-6	ES acc	ES $\bar{k}$	Ada acc	Ada $\bar{k}$	COMB-12
Qwen3.8-Max	93.3	93.3	93.3	93.3	2.0	93.3	4.1	93.3
Qwen3.8-Omni	93.0	93.9	93.0	93.0	2.2	91.5	4.3	93.0
Agnes3.0-Flash	78.8	89.4	90.9	90.9	2.5	90.9	4.6	90.9
Agnes2.5-Flash	90.0	94.2	95.0	95.0	2.1	92.5	4.2	95.0
GPT-6-Luna	80.0	76.7	73.3	80.0	2.0	86.7	4.5	86.7
Gemini3.8-Flash	91.7	86.1	83.3	83.3	2.7	91.7	4.7	91.7

sufficed. Both simulations are exact replays of the released call matrices, so they upper-bound nothing and can be re-derived from the logs.

6. Discussion

6.1. Principal findings

The study yields four principal findings. First, the effectiveness of every test-time aggregation strategy is governed by a precondition that the literature rarely states explicitly: the model must produce diverse answers at all. For the flagship models in our pool, five of six benchmarks sit in a degenerate regime in which six stochastic samples and six paraphrases return the same answer for almost every question, and in that regime self-consistency, paraphrase ensembling, and the combined ensemble reproduce the greedy baseline exactly, with the hybrid allocation differing by at most one question. This part of the degeneracy result is structural rather than empirical: when all votes coincide, every aggregation of them returns the same answer by construction. The empirical content lies in how common the coincidence is, and in its consequence that agreement-based uncertainty scores are uninformative exactly there. Second, in the high-entropy regime, which in our suite is concentrated in logical reasoning for weaker models and in selected quantitative tasks for mid-tier models, both pure strategies improve over greedy decoding when the correct answer is the modal one, while the highest-entropy cells show that majority voting can leave accuracy unchanged or even reduce it when the modal answer is wrong, in line with the plurality-vote model of Section 3.7. Third, paraphrase agreement functions as an error signal in its own right, and its error coverage overlaps only partially with that of sample agreement: across models, the stronger source differs, so neither signal subsumes

the other, and a deployment that can afford both probes obtains partially disjoint evidence, although the simple sum of the two disagreement scores does not dominate the best single signal in our data. Fourth, fragility, the fraction of questions whose answer flips under paraphrasing alone, tracks answer entropy, concentrates entirely in stem-locus paraphrasing, and appears in the same model-task cells that benefit from ensembling, which supports the interpretation that paraphrase disagreement and sampling disagreement are two observable symptoms of the same underlying instability.

6.2. A practical allocation rule, and how it performs

The regime structure suggests a simple procedure for practitioners with a fixed per-question budget. Probe the question first with a small number of calls, compute sample agreement and paraphrase agreement, and branch on the result. If both agreement scores equal one, additional calls of either kind are unlikely to change the answer, and the budget is better spent elsewhere; the degenerate regime is identifiable from a handful of calls at negligible cost. If sample agreement is high but paraphrase agreement is low, the model is prompt-brittle, and the remaining budget should go to paraphrases. If the reverse holds, the budget should go to resampling. If both scores are low, split the remaining budget between the two sources.

We evaluated this rule by simulation on the collected call matrices, charging the probe's four calls (two canonical samples and two paraphrases) to the adaptive condition's budget. The adaptive rule matches or exceeds every fixed six-call strategy for the models where the strategies differ at all: it returns 86.7% at a mean cost of 4.5 calls for GPT-6-Luna, against 73.3% for fixed SC-6, and 91.7% at 4.7 calls for Gemini3.8-Flash, while for the degenerate flagships it terminates after the probe at a mean of 4.1 calls with the same accuracy as any full ensemble. Across all six models, an even simpler sequential rule that draws canonical samples until the first unanimity and otherwise falls back to the six-sample majority preserves SC-6 accuracy within one question everywhere at a mean budget of two to three calls. Because agreement statistics are cheap to compute online, the rules add no model training and no human labeling, and both simulations are reproducible from the released logs. The hybrid allocation, by contrast, is best understood as the fixed-budget version of the same hedge: it matches the best pure strategy when one source fails and trails it by about one question when both work, so the adaptive probe dominates it in our data at two-thirds of the cost.

6.3. Implications for evaluation and for uncertainty estimation

Two methodological consequences follow. For benchmark evaluation, our results add an aggregation dimension to the multi-prompt argument of Mizrahi et al. [1]: the same question answered under one prompt with one decode, under six paraphrases, or under twelve combined calls yields systematically different accuracies in the high-entropy regime, so reported scores should state the aggregation configuration, and comparisons across papers that aggregate differently are not like for like. The single-sample baseline S1 deserves the same reporting discipline, because greedy decoding and a single stochastic draw differ by up to ten points in the high-entropy cells of our data, in both directions. For uncertainty estimation, the degenerate regime is a warning whose scope we state precisely. Our measurements cover multiple-choice questions answered directly, where sampling-based confidence measures, including simple agreement heuristics, are uninformative when the sample distribution collapses, and our data show that this collapse is the common case for strong models on saturated tasks. Semantic-entropy methods for free-form generation [6, 7] face an analogous risk when the sampled paraphrase classes collapse, but our design does not test that setting, and the caution transfers by argument rather than by measurement. Paraphrase probes partially restore the signal because they perturb the input rather than the decoder, but they too inherit the collapse when the model is robust in both views, as our flagship models are on five of six benchmarks. Confidence methods should therefore be validated per regime rather than averaged over mixed test sets, and deployments should verify that the answer distribution carries variance before trusting agreement-based scores.

6.4. Limitations

Eight limitations bound the scope of our conclusions. First, sample size and power: fifteen questions per benchmark and completed sets of 12 to 90 questions per model support paired within-model comparisons of large effects, but the cluster-bootstrap intervals are wide, TOST does not establish equivalence for most pairs, and differences of a few points between strategies would go undetected; scaling each benchmark to hundreds of questions is the single change most likely to sharpen the conclusions. Second, provider rate limits and free-tier quotas constrained coverage unevenly across models, and the quota of Qwen3.8-Omni was exhausted before its follow-up conditions could be run; coverage follows a fixed question order rather than adaptive selection, but cross-model aggregates weight models unequally. Third,

the models are commercial systems accessed through third-party endpoints: we cannot verify that the requested temperature is honored, especially for reasoning-style models, and decoding internals, post-training, and version pinning are outside our control; findings describe the accessed systems as of October 2026 and need not transfer to other versions or serving configurations, although we release per-call timestamps and token usage so the configuration is auditable. Fourth, paraphrases were generated by two models and validated by embedding similarity with a fixed threshold plus a bidirectional entailment check, but semantic equivalence of adversarial questions is not fully formalizable, and stricter thresholds or human-curated paraphrases could shift the fragility statistics. Fifth, the multiple-choice tasks use direct answering while GSM8K uses chain of thought; the aggregation comparison is internally valid within each answering mode, and the follow-up condition of Section 5.8 measures the mode effect directly, but absolute accuracies are not comparable across modes. Sixth, the high-entropy regime is sampled through fewer model-task cells than the degenerate regime, resting primarily on logical reasoning and selected quantitative tasks; the MMLU-Pro probe of Section 5.7 begins to broaden this, at small scale. Seventh, error-detection metrics are computed against ensemble references that share votes with the detectors; we separate the definitional degenerate stratum and report disjoint-vote correlations, but an error reference from fully independent draws would be cleaner. Eighth, all conditions use a fixed budget of six or twelve calls with majority voting; ranked voting, weighted aggregation, or learned vote combination could change the budget-matched comparison.

6.5. Future work

Three extensions follow naturally. The adaptive allocation rule can be moved online, interleaving probing and allocation per question rather than simulating the split afterwards, and its threshold can be tuned against a target budget rather than fixed at unanimity. The comparison can be extended to free-form generation, where semantic clustering replaces majority voting and paraphrase ensembles interact with surface-form competition effects [40], and where the degeneracy phenomenon has direct consequences for semantic-entropy hallucination detection that our multiple-choice design cannot measure. Finally, the fragility statistic can serve as a benchmark diagnostic in its own right, complementing accuracy with a measure of how strongly a model's answers depend on the surface form of the question.

7. Conclusion

We presented a budget-matched comparison of the two diversity sources available to test-time ensembling for large language models: resampling the decoder on a fixed prompt, and perturbing the prompt across validated semantic paraphrases. A sixteen-call matrix per question per model, evaluated on six models from four families and six benchmarks, allowed every condition, including self-consistency, paraphrase ensembling, a hybrid allocation, and a combined ensemble, to be derived from matched calls on identical questions, and follow-up controls isolated the prompt-space contribution with greedy paraphrase decodes, probed harder tasks on MMLU-Pro, and varied the answering mode with chain-of-thought instructions. The results organize into a regime structure. Where models are confident, answer distributions collapse, all aggregation strategies coincide with greedy decoding, and consistency-based uncertainty estimation loses its signal. Where answer entropy exists, both diversity sources improve accuracy, paraphrase agreement flags errors that sample agreement misses and vice versa, and the hybrid allocation hedges: it matches the best pure strategy when one source fails, at a cost of about one question when both work. No significant difference between the two diversity sources was detected at matched budgets, and the study is explicit that its sample sizes do not establish equivalence. The practical takeaway is to measure agreement first and allocate the budget accordingly: an adaptive rule simulated on the collected matrices matches or exceeds every fixed strategy at two-thirds of the budget, and a sequential unanimity rule preserves self-consistency accuracy at a mean cost of two to three calls. The methodological takeaway is that aggregation configuration, and the choice between greedy and single-sample baselines, belong in the reporting standard of every evaluation that uses them. We release prompts, paraphrases, and raw predictions to support replication as model behavior continues to shift.

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